3.4 Derivatives of Logarithmic Functions

Name _____ Period ____ Date _____

AP Calculus 3.4 (30-minute plan) · Student sheet

I can…

- find the derivatives of ln x, ln u, and log_b x.
- find the derivative of b^x.
- use logarithmic differentiation, including when the variable is in the base and the exponent.

Vocabulary · fill in the missing word

Derivative of ln x — $\frac{d}{dx}$ ln x = _____, x > 0

Derivative of log_b x — $\frac{d}{dx}$ log_b x = _____

Logarithmic differentiation — take ln of both sides, use log _____, differentiate implicitly, then solve for y′

e as a limit — e = _____

Part 1 · Notes and examples (fill in each note, then work the example)

Natural log: d/dx ln x = 1/x, and d/dx ln u = _____.

Example 1

Find each derivative.

a. $y = \ln(x^2 + 5)$

b. $y = x \ln x$

Rewrite first: use log properties so that ln of a product or quotient becomes a _____ or difference of logs.

Other bases: d/dx log_b x = 1/(x ln b) and d/dx b^x = _____.

Example 2

Find each derivative.

a. $y = \log_2 x$

b. $y = 5^x$

Logarithmic differentiation: take ln of both sides, differentiate, then multiply by _____.

e as a limit: the limit as n → ∞ of $(1 + a/n)^n$ = _____.

Example 3

Find y' for $y = x^x$, $x > 0$.

Part 2 · Partner practice

Partner practice

(AP® Practice) If $h(x) = \ln(x^2 + 4)$, then $h'(x)$ equals

(A) $\frac{2x}{x^2 + 4}$ (B) $\frac{1}{x^2 + 4}$ (C) $\frac{x^2}{x^2 + 4}$ (D) $\left|\frac{2x}{x^2 + 4}\right|$